

Enhancing Transparency in Large Language Models to Meet EU AI Act Requirements

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Abstract

Transparency is a quintessential requirement in artificial intelligence systems, featuring prominently under the EU AI Act, but it raises immense challenges for LLM developers due to the intrinsic black-box nature of LLMs. The paper discusses some techniques and methodologies that could assist in increasing the transparency of LLMs and achieving conformity with the exacting standards prescribed under the EU AI Act. In this paper, the authors go further in developing the demands that the Act places on transparency, documentation, and human oversight. Moreover, there are inherent issues associated with interpreting LLMs, particularly the trade-offs between performance and interpretability, compounded by intricate systems. Additionally, this paper gives an overview of the current state of the art in explanation methods for LLM outputs, ranging from attention mechanisms over model distillation to post hoc explanation tools. Some guidelines that developers should follow include the following: performance vs. interpretability trade-offs, full documentation, and user-centered design principles. Through illustrative case study, we demonstrate how these practices can be implemented in GPT-based models to meet regulatory demands. By bridging the gap between advanced AI models and the EU's regulatory framework, this paper contributes to the development of AI systems that are not only powerful but also transparent, fostering greater trust and acceptance among users and regulators.

Keywords

Large Language Models, EU AI Act, Transparency, Neural Networks, Artificial Intelligence



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1 Introduction

In fact, LLMs have been storming the NLP world in recent times: this new breed attained unparalleled capability in such wide activities as machine translation, summarization, and generation of dialogues, among many others. Famous examples include: BERT [11], GPT-4 [30], LLaMA 3.2 [27], among many more that assure state-of-the-art results by combining the power of deep neural network architectures with very large datasets. Despite these successes, such models often receive heavy criticism because of their "black-box" nature, where decision-making processes are not transparent or interpretable [25]. This lack of transparency raises many concerns about accountability, trust, and compliance with emerging regulations.

In this respect, the European Union has addressed these concerns by introducing the AI Act, which is aimed at developing a legal framework that would make AI systems transparent, explainable, and consistent with basic rights [13]. It categorizes AI applications by risk level and then lays extremely high demands on high-risk systems. As it has already been mentioned [24] the efficacy of contemporary cybersecurity technologies and procedures has not attained the requisite maturity to safeguard high-risk AI Systems against threats. Particularly, in the case of LLMs, the AI Act does not classify them as a high-risk AI system [13], but it becomes crystal clear that it can equally be a source of high risk, depending on how it would have been implemented in the organization process, provided the condition that it is used and the use of its outputs [14]. Key commitments involve transparency provisions, documentation requirements, and the need for explanations on AI-driven decisions [36].

Meeting these regulatory demands creates significant challenges for the developers of LLMs. The interpretability problem of a neural network, typically containing millions or billions of parameters, is inherently intractable when trying to explain how inputs are transformed to yield a given output [34]. Often, increasing interpretability comes at the cost of sacrificing performance since simpler models lack the representational capacity for capturing the complex patterns learned by deeper models [12].

Several techniques have been proposed to make artificial intelligence models more transparent and easier to understand. Attention mechanisms, integral to models like Transformers, help focus on input aspects contributing to the model's predictions [10]. Model distillation is the process of converting complex models into more understandable ones without reducing their performance [22]. Various post hoc explanation techniques like LIME [33] and SHAP [26] provide local-level explanations for the specific predictions made by the models, thereby enabling a better understanding of model behavior.

This paper undertakes an investigation of such methods from the point of view of compliance with the EU AI Act. We will discuss in detail the legal requirements for transparency, analyze the intrinsic problem associated with the interpretation of LLMs in this work, further review the current methodologies for enhancing interpretability, and finally put forward some practices that developers should follow to enhance and improve the transparency of LLMs without influencing their performance. The goal is to enhance trust and adoption for both users and regulatory bodies by aligning technological advancements with regulatory requirements.

The rest of the paper is organized as follows: Section 2 reviews the relevant legal obligations under the EU AI Act; Section 3 explores techniques to enhance transparency; Section 4 lists techniques for developers; Section 5 presents case study showing in practice how such techniques may be implemented; and finally, Sections 6 & 7 concludes the paper with suggested directions for future research.

2 Legal Obligations Under the EU AI Act

The most important step forward for its regulation has been made with the proposal of the EU AI Act, which aimed at ensuring that those systems are developed and used in ways that respect ethical principles and foster transparency in addition to adhering to basic rights [13]. To such a purpose, this law has established a sound framework for legally classifying AI systems, with regard to their associated risks, and for imposing corresponding duties on providers and users [36]. ANNEX XIII of the EU AI Act delineates the criteria for the classification of general-purpose AI models associated with systemic risk, as referenced in Article 51. The criteria are:

- (a) The number of parameters in the model;
- (b) the quality or magnitude of the data collection, for instance, quantified by tokens;
- (c) the computational resources used for model training quantified in floating point operations or represented by a mix of factors such as projected training cost, anticipated training duration, or predicted energy expenditure during training;
- (d) the model's input and output modalities, including text-to-text (big language models), text-to-image, multi-modality, and the

current benchmarks for assessing high-impact capabilities for each modality, as well as the particular types of inputs and outputs (e.g., biological sequences);

(e) the benchmarks and assessments of the model's capabilities, encompassing the number of tasks performed without supplementary training, its adaptability to acquire new, unique tasks, its degree of autonomy and scalability, as well as the tools available to it;

(f) if it significantly affects the internal market owing to its scope, which will be inferred when it is accessible to a minimum of 10,000 registered business users located inside the Union;

(g) the number of registered end-users.

3 Transparency Requirements

The notion of transparency is fundamental to the EU AI Act, signifying the necessity for explainability and accountability in artificial intelligence systems [19]. The Act stipulates that Generative AI, such as LLMs, comply with transparency requirements and EU copyright laws. 1) Disclosing that the knowledge was generated by artificial intelligence 2) Formulating the model to prevent the creation of illicit content. 3) Distributing summaries of copyrighted content utilized for teaching purposes.

High-impact general-purpose AI models, such as the sophisticated GPT-4, that may present systemic risks must undergo comprehensive examinations, and any significant events must be notified to the European Commission. Content that is either produced or altered with the assistance of AI—such as photos, audio, or video files—must be clearly labeled as AI-generated to ensure consumers are informed when encountering such material.

3.1 Human Oversight

The EU AI Act has stated that human oversight shall be endowed in order to avoid the unstoppable operations of AI systems, enabling interference where it calls for it [13]. According to Article 14, AI systems have to be designed and developed in such a way to provide human oversight, including:

- **Human Interface:** Allows the user or operator to amend or revoke a decision taken by the AI system [2].
- **Monitoring Mechanisms:** Providing tools for humans to monitor the AI system's performance and detect anomalies or malfunctions [39].
- **Training and Support:** Ensuring users have sufficient training to engage with the AI system and comprehend its results [1].

That is, in LLM applications, implementing moderation tools should go along with providing interfaces to feedback and correction, so that the understanding and management of AI behavior by users becomes possible.

3.2 Developer implication in AI

The AI Act lays down requirements whose direct effect on the developers of AI, especially regarding LLMs, is grave:

- **Improved Development Efforts:** Transparency and human intervention mechanisms demand more funds to be implemented. [41].

- **Need for Interdisciplinary Expertise:** Developers may require collaboration with legal experts, ethicists, and domain specialists to assure compliance. [15].
- **Compliance Costs:** Documentation, risk assessments, and continuous compliance works may have financial implications - costs. [7].
- **Innovation Challenges:** Legal compliance and innovation are at odds, as meeting the strict standards will limit the team's flexibility in introducing new features and swiftly deploying AI systems. [9].

Clear implications herein will ease the way developers get along with regulatory requirements. Given its early integration into the development framework, AI developers should be prepared to address legal requirements and build trust with users and stakeholders.

4 Techniques for Developers

The creators of Large Language Models (LLMs) face the dual responsibility of advancing the field of natural language processing while complying with regulatory standards such as the EU AI Act. Adopting useful practices that prioritize transparency is crucial not only for meeting legal obligations but also for fostering user trust[41]. This section outlines various strategies that developers can employ to enhance the interpretability of LLMs with minimal impact on their overall performance.

4.1 Performance aligned with Interpretability

It is considerably hard to strike a balance between high-level performance and interpretability [12]. Complex models that are usually the best in performance are rarely interpretable. It is left to the developers to make a compromise in which the model is effective but at the same time transparent enough to satisfy regulatory and user demands. Approaches that may help in balancing the trade-off between performance and interpretability:

The techniques that reduce the models' complexity include model pruning or distillation with minimal loss of performance. [22].

- **Hybrid Models:** Combine black-box models with interpretable ones. The latter give an explanation for decisions taken by the former [5].
- **Accommodating the performance metric:** Interpretability metrics will be implemented within the model evaluation framework alongside the existing set of traditional performance measures. [18].

Through good trade-offs among these various factors, developers can come up with models that are efficient yet meet some standards of transparency.

4.2 Documentation and Transparency

Comprehensive documentation undergirds the transparency requirements for the EU AI Act [13]. What is needed is detailed documentation to be provided by developers on the working mechanics of LLMs: the data they use, and the limitations or intrinsic biases therein. Optimal practices for documentation include:

- **Model Cards:** Model cards for consistently documenting model specifications, performance attributes, and ethical implications. [28].
- **Dataset Sheets:** Provide dataset sheets filling in the details of datasets pre-processed for training, sources, collection methods, and pre-processing applied to the datasets. [16].
- **Transparency Reports:** Releasing transparency reports explaining the objectives of the AI system, design decisions, and compliance procedures. [32].
- **Regression Updates:** One should not forget documentation to reflect changes made in the model, or the environment where it is implemented. [31].

Comprehensive documentation facilitates compliance and bolsters user comprehension and confidence in AI systems.

4.3 Ethical Considerations and Bias Mitigation

Addressing Ethical Concerns and Possible Biases: LLMs are of utmost importance from a compliance and social responsibility perspective. Developers should make efforts to profess and eliminate biases that might exist either in training data or model behaviors. Optimum methodologies would include:

- **Bias Audits:** Conduct structured analyses to identify partiality in the model outputs and implement mitigating controls as necessary [6].
- **Diverse Training Data:** Training on diverse and representative data reduces the potential for biased results in models [42].
- **Fairness criteria:** Use fairness criteria to benchmark and monitor model performance against different demographics [37].
- **Stakeholder Consultation:** Have an understanding of obvious implications from stakeholders, particularly those from communities that could be impacted, and obtain feedback. [20].

Addressing ethical issues well in advance allows developers to make the LLM more acceptable and credible.

4.4 Continuous Improvement and Monitoring

Compliance and transparency are ongoing processes that require constant monitoring and upgrading [3]. Developers should provide mechanisms for checking model effectiveness and updating models if necessary. The following suggestions are recommended:

- **Monitoring Systems:** Design and implement monitoring mechanisms to surveil model outputs, performance metrics, and user input in real-time [35].
- **Periodic Updates:** Revising regularly by updating with the latest data available, correcting problems found, and enhancing performance. [4].
- **Incident Response Plans:** Develop procedures for responding as quickly as possible to any adverse event or malfunction of the AI system [3].
- **Compliance Audits:** Occasionally check compliance with regulatory requirements, and update the process to match constant changes in legislation and regulations [7].

Continuous improvement ensures that over time, LLMs prove to be effective, compliant, and meet users' needs.

5 Case Study

This case study would provide evidence for some of how LLMs would align with the EU AI Act in giving clarity to increased transparency by way of records to be kept for compliance with applicable laws and regulations.

5.1 Compliance Case Studies under EU AI Act

This case study, therefore, shows how developers could stick to ethical principles scrupulously and keep those records that may help bring more transparency and conformance to the GPT-based model in the EU AI Act.

5.1.1 Model Cards and Data Sheets Development. The developers of the GPT-based model followed exactly the recommendations of Mitchell et al. [28] and Gebru et al. [16], along the whole course of its development. Especially the documentation contained:

- **Model Specifications:** Model architecture, goals of training, and versioning to track which model versions are being used.
- **Training Data:** Data description, collection practices, and possible biases.
- **Evaluation Metrics:** The evaluation indicators used on different tasks and standards.
- **Statement of Purpose:** Do's and Don'ts in detail as instructed by the model.
- **Ethical Issues:** Discussion on possible risks; amongst others, the generation of noxious substances shall be considered and mitigated.

By providing this information, developers were following the EU AI Act's needs for transparency and disclosure to allow users and regulators to understand the model's capabilities and limitations.

5.1.2 Performing Risk Assessments. The developers ran extensive risk analyses in order to find and minimize all possible model-related issues [21]. This included:

- **Bias Analysis:** Checking the model for various biases regarding gender, race, and other protected attributes using well-established benchmark datasets [23].
- **Adversarial Robustness:** Evaluating the model for its potential susceptibility to adversarial inputs that can result in wrong outputs [40].
- **Privacy Considerations:** Protection against the unauthorized leakage of sensitive personal information from the training dataset [8].

Following the evaluations, developers instituted measures to mitigate identified issues, which included refining the model via bias-reduction methods [42] and integrating filters designed to identify and avert harmful content. [17].

5.1.3 Promoting Human Oversight and User Feedback. Consistent with the emphasis on human governance articulated in the EU AI Act, engineers have developed the system to integrate:

- **User Controls:** Features allowing users to choose between more creative or random content.

- **Feedback Mechanisms:** An intuitive interface that enables users to report problems or unsatisfactory outcomes, thereby promoting ongoing enhancement [2].
- **Transparency Notices:** Informing users about their interaction with an AI system and providing guidance on how to use it appropriately.

These methods involve the user in making better decisions and maintaining control of the outputs from the AI system.

5.1.4 Development of Compliance Policies. Developers instituted internal structures that would ensure constant compliance with the EU AI Act:

- **Auditing:** Routine, structured assessment of model performance, equity, and safety [32].
- **Legal Consultation:** Engaging with legal experts to examine regulatory requirements and adjust practices accordingly [38].
- **Stakeholder Engagement:** Informed and eliciting the views from users, ethicists, and concerned communities on concerns and issues. [29].

By including such methodology, the developers basically indicate that they are committed to ethics in developing AI and also follow regulatory benchmarks.

5.1.5 Outcomes and Gained Insights. Applying Compliance Protocols led to:

- **More Trust:** Participants have gained more trust in the AI system because such transparency in understanding the model outcomes lent more confidence.
- **Enhanced Performance:** Feedback mechanisms made models constantly improve for better performance.
- **Compliance with Regulations:** The plan ensured that the AI system was within the limits of the EU AI Act for legality.

This case study shows that investment in compliance not only meets legal obligations but also offers both commercial and ethical benefits.

6 Discussion

Improving LLM transparency is complex, yet important work toward the compliance of the EU AI Act. This research studied the current landscape, underlining the fact that any exploitation of large language models' extraordinary capabilities will need to overcome serious technical hurdles and adhere to an ethical ecosystem to meet regulatory compliance. While GPT-4 and LLaMA 3.2 did an outstanding job of performing natural language processing tasks, there is still a serious lack of interpretability because of "black-box" features, again raising many questions related to the issues of trust and accountability. Aggravated are the current proposals before the EU AI Act, with its paradigm on artificial intelligence systems laid down within the principles of transparency and human supervision. This ongoing debate on various approaches-attention mechanisms and model distillation among others-keeps pointing out that yes, there are indeed workable technical alternatives toward making LLMs more explainable. However, every approach also carries its own set of trade-offs: attention methods may reveal some input elements impacting a model's prediction, but these

rarely create much in-depth holistic insight into how the model's general working mechanisms function. Similarly, model distillation might offer better interpretability for complex models; this, however, may be at the expense of reduced performance by the overall model or reduced capability to spot complex patterns. Anyway, developers will need to consider the trade-offs in performance and interpretability in light of the strict requirements of the EU AI Act. The case study presented within this paper provides novel insights into how there could be implementation of transparency within AI systems. The development of model cards, data sheets, and compliance regulations is a novel example of a framework for developers willing to enable comparability of transparency together with performance achievement. Risk assessments - realized by audits about bias and adversarial robustness - are needed to confirm large language models operate within regulatory requirements, but also to address larger social concerns about equity and the ethics of using AI. The techniques that we mentioned, highlight that ongoing monitoring and improvement emphasize the ever-changing nature of AI systems. These techniques can offer a baseline for secure development and deployment of AI systems. This baseline may serve as the foundation for developing an AI security framework. Compliance is indeed an ongoing process in which the developer, periodically and critically, needs to review and make improvements in their systems in light of emerging data, threats, and changes in legislation. Only such proactive action would instill confidence among users and regulators, amongst others, that the LLMs are reliable and do not contravene evolving standards of ethics. The future of LLMs depends on how developers are able to infuse transparency and accountability mechanisms into such models without compromising the innovative potential of these models. Only close collaboration between engineers, ethicists, legal experts, and policymakers can address the challenges thrown up by the EU AI Act and other international regulatory frameworks.

7 Conclusion

This work discussed several challenges and approaches towards making LLMs transparent under the EU AI Act. We had the opportunity to show, through reviews of several state-of-the-art methodologies using attention mechanisms and model distillation, that methods leading to improved interpretability are indeed possible but that developers have to carefully balance these against trade-offs in performance. They are bound to implement optimum methodologies that include comprehensive documentation, bias assessment, and ongoing oversight, among others, for regulatory requirement assurances coupled with user confidence. This case study is a good example of how pragmatic strategies such as model cards and risk assessments feature directly within the workflow of AI development. Setting aside their legal necessity, these types of strategies have an inherent benefit in improving ethics within AI systems. Future work-better methods of explainability, better interdisciplinary collaboration, and standardization - will make sure emerging projects meet compliance with the EU AI Act and other legislation as large language models continue to evolve. This makes the development life cycle of LLMs uniquely transparent and accountable - an opportunity for the AI community to focus on developing robustly

moral yet powerful systems that would enjoy greater acceptability from users, regulators, and society in general.

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